Twists as auxiliary trace fibers

2026-11-13 PR3560 Source-only desk derivation

Status. Proposed identities, unelaborated Lean source and next-test design. No scientific execution, checked theorem, IDC1 or measured improvement. All research assertions remain `[pcc:unverified]`.

Question. Can retaining the twist coordinates of a trace fiber deliver useful source relations with lower certified solver degree or work? The comparison must give the original curve the same extended formulation.

Take nonsingular $E/\mathbb{F}_q : y^2 = x^3 + Ax + B$, characteristic at least 5, d nonsquare, $K = \mathbb{F}_q(w)$, $w^2 = d$, and actual $Q = (a, b) \neq O$. The line $L(X) = \lambda(X - a) - b$ has residual quadratic

$$q_\lambda(X) = X^2 - sX + p, \qquad s = \lambda^2 - a, \quad p = a\lambda^2 + 2b\lambda + a^2 + A.$$

The proposed nonsplit chart is

$$\boxed{dt^2 = \Delta_Q(\lambda) = \lambda^4 - 6a\lambda^2 - 8b\lambda - 3a^2 - 4A.}$$

For $t \neq 0$, $X = (s + wt)/2$ and $Y = L(X)$ give conjugate candidates $R, \sigma R$ on E with $R + \sigma R = Q$. These are extension points, *not* two rational source factor-base points. For $t = 0$, retain the tangent condition $2R = Q$ separately. The identity query needs its own branch.

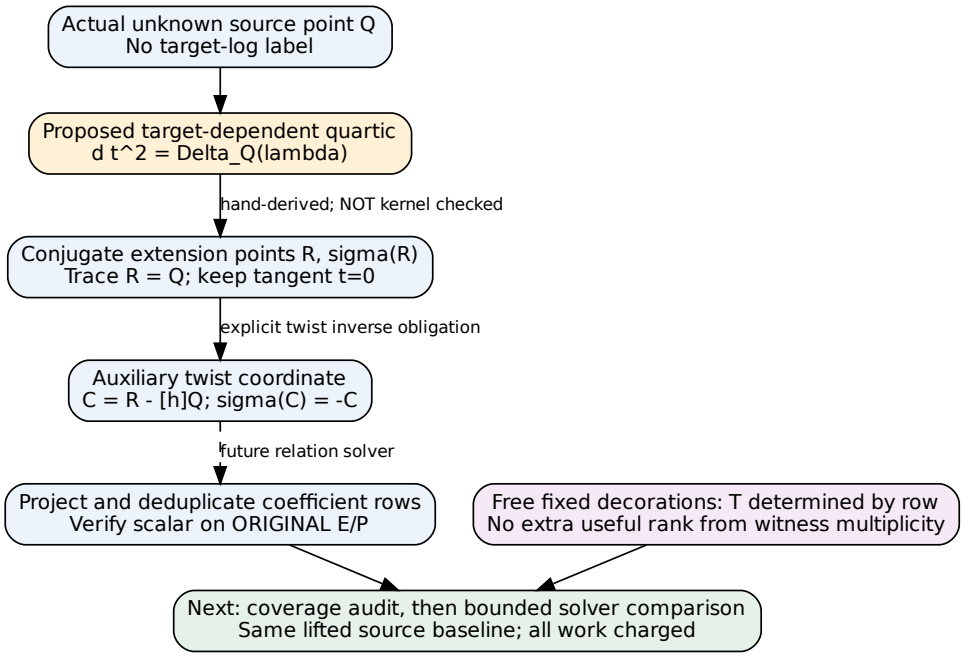


Diagram: proposed routes only. No numerical values or measured edges.

Source-preserving interpretation

For the monic twist $T : Y^2 = X^3 + d^2AX + d^3B$ use $\psi(X, Y) = (X/d, Y/(dw))$, with infinity handled separately. Let Q lie in an explicitly supplied odd-order subgroup of order r, and choose public h with $2h = 1 \pmod r$. Translation by $S_Q = [h]Q$ identifies a candidate trace fiber with the Frobenius-minus group:

$$C = R - S_Q, \qquad \sigma C = -C, \qquad R = S_Q + \psi(T_{\text{aux}}).$$

The inverse coordinates $(dx(C), dwy(C))$ are proposed to be rational over $\mathbb{F}_q$. This is a twist torsor/presentation, not an assumed rational main-to-twist group isomorphism. No unknown target scalar is used.

Derivation obligations. Expand $s^2 - 4p$, then use $4(X^2 - sX + p) = (2X - s)^2 - (s^2 - 4p)$. Combine the quadratic root condition with the existing curve/line factorization. Actual Frobenius and the nonsingular chord law must still be bound formally. The four handwritten algebra declarations in `drafts/TraceFiber.lean.txt` have not run. Tangent, projective and twist-inverse coverage remain explicit open obligations; no IDC1 is issued.

Control: a free twist witness is not a new row

For fixed $A_i = [h]B_i + \psi(T_i)$, a valid source relation implies

$$\sum_i c_i A_i = [h]Q + \psi\left(\sum_i c_i T_i\right).$$

Taking trace gives the converse source relation; injectivity of ψ makes the auxiliary witness on a valid row exactly $\sum_i c_i T_i$. Thus fixed free decorations cannot create additional complete projected coefficient rows. The remaining hypothesis concerns a solver's certificate degree/delivery cost. Restricting the witness can lose rows; varying lifts requires explicit membership and deduplication before rank.

Next discriminating test

First implement independent Rust/C++ chart and point-reference instruments for historical $\mathbb{F}_{37}$ inputs $E = (7, 19)$, $d = 35$, $T = (28, 33)$, $P = (1, 8)$, $r = 19$, $h = 10$. Test every chart pair for all 18 nonidentity source points, retaining $t = 0$, signs and a separate identity case. There are $18 \cdot 37^2 = 24{,}642$ *planned* candidate checks, not distinct DLPs. No instrument, census or new recovery exists yet.

After coverage, separately freeze the complete source-policy/frontier comparison, legitimate cold public-log setup and original-point scalar recovery. Charge all extension, map, root, failed-query, rank and setup work. Track actual matrix degree and expanded original-input witness degree; quartic input degree alone is not regularity or speed. Preserve every loss. Each stage is capped at 600 seconds, two existing local workers and one window at most 3600 seconds, after a successful approved official checkpoint.

Sources checked. Gorla–Massierer, section 2.2 for trace/Frobenius (their algorithmic analysis is for $n \geq 3$); Kosters–Yeo, sections 2, 4–5 for summation systems, descent and degree cautions; existing repository `IDEAS.md`/chord design and Nov12 leads. KB MCP unavailable; official wrapper returned127. No novelty claim or signed verification.

Execution. Fresh runtime lacks old local caches and narrow token. Published sources were restored; no proof/native experiment or S3 upload ran. Old frozen studies, identities, graphs, metrics and ledger history are unchanged.